

IB MATH AA SL

Lesson 38

Stationary Points & Curve Sketching

Calculus · 60-minute lesson

Syllabus SL 5.7–5.8 · Property of Lynda Badmus Education

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Do Now — recall

ON YOUR WHITEBOARD

1. Differentiate $y = x^2 - 4x$.
2. Solve $2x - 4 = 0$.
3. At a 'turning point', what is the gradient?

Answers (click to reveal)

1. $2x - 4$
2. $x = 2$
3. zero

What we're learning today

LEARNING OBJECTIVES

- 1 Find**
stationary points by solving $f'(x) = 0$.
- 2 Classify**
them as maxima or minima.
- 3 Use**
the second derivative test.
- 4 Sketch**
a curve using this information.

SUCCESS CRITERIA — I CAN...

- ✓ solve $f'(x) = 0$ to find stationary points
- ✓ find the y-coordinate of each point
- ✓ use $f''(x)$ to classify max/min
- ✓ sketch the curve's shape

Stationary points

WHERE $f'(x) = 0$

$$f'(x) = 0$$

Stationary

the gradient is zero (a flat point)

Find x

solve $f'(x) = 0$

Find y

substitute back into the curve

TYPES

Maximum

peak — curve turns down

Minimum

trough — curve turns up

Inflexion

a flat point that is neither.

Find stationary points

QUESTION

Find the stationary point of $y = x^2 - 4x + 1$.

SOLUTION

Differentiate and set to zero:

$$\frac{d y}{d x} = 2 x - 4 = 0$$

Solve: $x = 2$. Then $y = 2^2 - 4(2) + 1$:

$$y = -3$$

Stationary point: $(2, -3)$.

Your turn — stationary

ANSWER IN YOUR BOOK

1. $y = x^2 - 6x$. Find x at the stationary point.
2. $y = x^2 + 2x$. Find x at the stationary point.
3. Solve $dy/dx = 3x^2 - 12 = 0$.

Answers (click to reveal)

1. $x = 3$
2. $x = -1$
3. $x = \pm 2$

The second derivative test

MAX OR MIN?

$$f''(x) < 0 \Rightarrow \text{max}, \quad f''(x) > 0 \Rightarrow \text{min}$$

Differentiate again

find $f''(x)$

$$f'' < 0$$

maximum (curve concave down)

$$f'' > 0$$

minimum (curve concave up)

SKETCHING

Plot stationary points + intercepts.

Shape

positive $x^2 \rightarrow \cup$; negative $x^2 \rightarrow \cap$.

Ends

follow the highest power.

Classify a stationary point

QUESTION

For $y = x^2 - 4x + 1$, classify the stationary point at $x = 2$.

SOLUTION

Second derivative:

$$\frac{d^2 y}{d x^2} = 2$$

$f'' = 2 > 0$, so it is a MINIMUM.

(Matches the U shape of a positive x^2 .)

Exam-style question

Consider $y = x^3 - 3x$.

- (a) Find dy/dx . [1]
- (b) Find the x -coordinates of the stationary points. [3]
- (c) Use $f''(x)$ to classify them. [2]

MARK SCHEME — reveal after students attempt (tutor only)

- (a) $3x^2 - 3$ (A1)
- (b) $3x^2 - 3 = 0 \rightarrow x = \pm 1$ (M1)(A1)(A1)
- (c) $f'' = 6x$; $x=1$: $f''=6>0$ min; $x=-1$: $f''=-6<0$ max (A1)(A1)

Common mistakes

**Set $f' = 0$**

Stationary points come from $f'(x) = 0$, not $f(x) = 0$.

**Find y too**

Give both coordinates of the point.

**Second derivative test**

$f'' < 0$ max, $f'' > 0$ min (easy to swap).

**Classify**

Don't leave a stationary point unclassified.

Mixed practice

QUESTIONS

1. $y = x^2 - 8x$. Stationary x ?
2. $y = x^2 + 4x + 1$. Stationary point?
3. $f''(x) = -4$: max or min?
4. $y = x^3 - 12x$. Stationary x 's?

Answers (click to reveal)

1. $x = 4$
2. $x = -2, y = -3 \rightarrow (-2, -3)$
3. maximum
4. $3x^2 - 12 = 0 \rightarrow x = \pm 2$

Plenary & exit ticket

KEY FORMULAS

$$f'(x) = 0$$

$$f''(x) > 0 \Rightarrow \text{min}$$

EXIT TICKET — before you leave

1. $y = x^2 - 2x$. Stationary x ?
2. $f''(x)=3$: max or min?
3. Gradient at a turning point?

Answers (click to reveal)

1. $x = 1$
2. minimum
3. 0

Homework

SET ON DR FROST MATHS (auto-marked)

Task:

Stationary points & curve sketching.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 39

Optimisation — using calculus to find best values.

$$\frac{dA}{dx} = 0$$